

Ph.D. Applied Qualifying Exam 2025 (Take Home)

July 10 (9 AM) to July 11 (5 PM)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. The exam consists of three problems, all of which you need to answer. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have questions, please email Prof Zhiqiang Tan (ztan@stat.rutgers.edu) for Problem 1 and Prof. Ying Hung (yhung@stat.rutgers.edu) for Problems 2 and 3, cc'ing Prof Tirthankar Dasgupta (td370@stat.rutgers.edu) and Prof Han Xiao (hxiao@stat.rutgers.edu).

Submission Instructions:

1. Create a folder containing the following:
 - (a) A word or pdf file containing your answers to the two problems, including all tables, figures and Appendices (this may be generated from an R markdown file).
 - (b) An R (or python) file containing the codes used to generate your results, or an R markdown file from which your output html or pdf is generated.
 - (c) A word or a text file that gives a summary of the codes submitted (not required if you submit an R markdown file).
2. Name the folder **yourlastname_yourfirstname_Appliedexam2025**. Zip this folder and email it to hxiao@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 5, Friday.

Problem 1

Let (X_i, Y_i) , $i = 1, \dots, n$, be a set of n observations, where X_i is a scalar covariate and Y_i is a scalar response. For fixed $\sigma > 0$ and $\lambda > 0$, consider

$$\hat{f} = \operatorname{argmin}_{b_0, b_1 \in \mathbb{R}, f \in \mathcal{H}_\sigma} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i - f(X_i))^2 + \lambda \|f\|^2 \right\}, \quad (1)$$

where $\mathcal{H}_\sigma$ is the reproducing kernel Hilbert space (RKHS) defined by the kernel

$$K_\sigma(x, z) = \exp\left(-\frac{(x - z)^2}{2\sigma^2}\right), \quad x, z \in \mathbb{R},$$

and $\|f\|$ is the norm of f in the RKHS.

1) Show that a solution in (1) can be represented as

$$\hat{f}(x) = b_0 + b_1 x + \sum_{i=1}^n c_i K_\sigma(X_i, x),$$

for some constants $b_0, b_1, c_1, \dots, c_n \in \mathbb{R}$, independent of x .

2) The coefficient vectors, $b = (b_0, b_1)^\top$ and $c = (c_1, \dots, c_n)^\top$, associated with $\hat{f}$, denoted as $\hat{b}$ and $\hat{c}$, can be determined as

$$(\hat{b}, \hat{c}) = \operatorname{argmin}_{b \in \mathbb{R}^2, c \in \mathbb{R}^n} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i - f(X_i; c))^2 + \lambda \operatorname{pen}(c) \right\},$$

where $f(x; c) = \sum_{i=1}^n c_i K_\sigma(X_i, x)$, and $\operatorname{pen}(c)$ is a penalty term. Give an explicit expression for $\operatorname{pen}(c)$, and derive a closed-form formula for $(\hat{b}, \hat{c})$.

3) Show that $\hat{f}$ is a linear smoother, i.e., the vector of fitted values $\hat{\mathbf{f}} = (\hat{f}(X_1), \dots, \hat{f}(X_n))^\top$ can be obtained as $\hat{\mathbf{f}} = \mathbf{S}\mathbf{Y}$ for an $n \times n$ matrix $\mathbf{S}$, independent of $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$. Give an explicit expression for $\mathbf{S}$.

4) Write R code to implement $\hat{f}$ in (1) using your formula for $(\hat{b}, \hat{c})$ and $\mathbf{S}$. For your code, include (σ, λ) as tuning parameters, and provide the degree of freedom ($\mathbf{df}$), defined as the trace of $\mathbf{S}$, as part of output.

5) Apply your R code to the ozone dataset (`Laozone.data`) as used in Stat 596, with the covariate `dpg` and response `ozone`. Please use $\sigma = 10$ and manually pick the choices of λ to achieve $\mathbf{df} = 4, 6, 8, 12$ up to the first decimal. For example, for $\mathbf{df} = 8$, your actual $\mathbf{df}$ should fall in $[7.95, 8.05)$. Draw your four fitted functions over `dpg` range $(-80, 120)$ along with the original data in a single figure. For $\mathbf{df} = 8$, report the fitted values $\hat{f}$ at the 5 smallest and 5 largest `dpg` values.

6) Repeat your application in Question 5, except with $\sigma = 40$.

Problem 2

A k-mean clustering with K prespecified:.... Assume n data points $X_1, \dots, X_n$, where $X_i = (X_{i1}, \dots, X_{ip})^T$. Denote the K groups by $A_1, \dots, A_K$ with centers denoted by $C_1, \dots, C_K$, where $C_k = (C_{k1}, \dots, C_{kp})^T$. A k -mean approach is to optimize:

$$\min_{A_k, C_k} \sum_{k=1}^K \sum_{x_i \in A_k} \|X_i - C_k\|^2,$$

where $\|\cdot\|$ is the standard Euclidean norm.

- (1). Design a simulation with three groups (the true K is 3) and $p = 10$ to demonstrate the performance of k-means clustering with different choices of K .
- (2). To perform k -means clustering while simultaneously identifying K , one approach is to modify the objective function to:

$$\min_{A_k, C_k} \frac{1}{n} \sum_{k=1}^K \sum_{x_i \in A_k} \|X_i - C_k\|^2 + \sum_{j=1}^p \lambda \|C_{(j)}\|,$$

where $C_{(j)} = (C_{1j}, \dots, C_{Kj})^T$. Proof the following equation:

$$\sum_{k=1}^K \sum_{x_i \in A_k} \|X_i - C_k\|^2 + \sum_{j=1}^p \lambda \|C_{(j)}\| = \sum_{j=1}^p \left(\frac{1}{n} (X_{(j)} - LC_{(j)})^T (X_{(j)} - LC_{(j)}) + \lambda \|C_{(j)}\| \right),$$

where $X_{(j)} = (X_{1j}, \dots, X_{nj})^T$ is the j -th variable across all sample points, and L is a cluster assignment matrix with $L_{ik} = \mathbf{1}(X_i \in A_k)$, $\mathbf{1}(\cdot)$ is an indicator function, $i = 1, \dots, n$, $k = 1, \dots, K$.

- (3). Based on (2), provide the pseudo code and analyze the same simulated data from (1). You may pre-specify the tuning parameter λ as a constant in your code.
- (4). Another approach is to assume that each observation X_i is drawn from a mixture model with $f(x) = \sum_{k=1}^K \pi_k f_k(x; \theta_k)$, where π_k is the mixture weight and $f_k(x; \theta_k)$ is a multivariate normal distribution. Using the same idea in (2), we can add a penalty term to the log-likelihood function for the estimation of unknown parameters in the mixture model:

$$\sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k f_k(X_i, \theta_k) \right) - n\lambda \sum_{j=1}^p \frac{|C_{(j)}|}{|\tilde{C}_{(j)}|},$$

where $\tilde{C}_{(1)}, \dots, \tilde{C}_{(p)}$ are the estimated cluster center from the standard k-mean clustering. Assume the cluster assignments, L_{ik} , are missing data (latent variables), and develop an EM algorithm to solve the problem. Please provide pseudo code with details for the E-step and M-step. Then, analyze the same simulation data from (1) using this method. You may pre-specify the tuning parameter λ as a constant in your code.

- (5). Compare the performance of the three methods and comment on the difference.

Problem 3

Assume you have to cut a very rare diamond. Before doing so, you want to ensure that the settings of your cutting machine are finely tuned to minimize the cutting force applied to the diamond. This can reduce power requirements, elastic distortion of the diamond, and tool wear. To do so, we first conduct a series of experiments. The first variable considered is the type of cutting tool, denoted by chamfer or hone, in the DIAMOND dataset. If chamfer is chosen, there are two additional variables, angle and length, required to be specified. The remaining variables are edge radius, rake angle, nose radius, cutting speed, feed, and depth of cut. The last column is the response variable, the cutting force.

- (1). First, ignore the columns “angle” and “length”. Based on the remaining seven variables, fit a Gaussian process model. Please describe your assumptions, the fitted model, and the estimated parameters. Then, calculate the mean squared prediction error based on the fitted model.
- (2). Based on this cutting process, which of the seven variables has the greatest effect on the cutting force? Please provide detailed reasoning.
- (3). Plot the predicted cutting force along with its pointwise 95% percent prediction confidence intervals as a function of the variable you identified in (2).
- (4). Based on model (1), find the optimal setting that minimizes the cutting force. Under this optimal setting, what is the predicted cutting force and its prediction uncertainty?
- (5). If we need to incorporate both “angle” and “length” and fit a single model based on these nine variables, what would be a reasonable choice of model? Please describe your model and the assumptions (It does not have to be a Gaussian process model, and you do not need to analyze the data.).